

CONGENITAL HEART DISEASE (Children)

Epidemiology

- Most childhood heart disease is **congenital**. **8 per 1000** liveborn have significant cardiac malformation; **30% need intervention in 1st year**.
- Minor abnormalities (e.g. bicuspid aortic valve) up to 53/1000.
- 9 most common anomalies = 80% of lesions; 10–15% complex; 10–15% have non-cardiac abnormality.

Atrial Septal Defect (ASD)

- Mostly sporadic. **Types**: ostium secundum (commonest, central septum/foramen ovale), ostium primum (= **partial AVSD**, with regurgitant left AV valve), sinus venosus.
- **Symptoms**: commonly **none**; recurrent chest infections/wheeze; **arrhythmias (4th decade+)**.
- **Signs**: loud S1, **wide & FIXED splitting of S2**, soft ejection systolic murmur (grade I–II); partial AVSD → apical pansystolic murmur (AV valve regurgitation).
- **Investigations**: CXR (cardiomegaly RA/RV, ↑pulmonary vascularity). **ECG**: **secundum = RBBB + right axis deviation; partial AVSD = SUPERIOR QRS axis** (negative in aVF). **Echo = mainstay**.
- **Treatment**: secundum → **catheter occlusion device** (surgery if large); partial AVSD/sinus venosus → surgery. Done **age 3–5 yrs** to prevent right HF + arrhythmias.

Ventricular Septal Defect (VSD)

- **Most common CHD (30%)**. Types by position (membranous, perimembranous, inlet, muscular) and size (small/large).
- **Small VSD**: usually asymptomatic; **loud pansystolic murmur at lower left sternal edge** (**loud = SMALLER defect**); quiet P2.
- **Large VSD**: dyspnoea, feeding difficulty, poor growth, sweating, recurrent chest infections, **HF in early infancy**; tachypnoea, hepatomegaly. Signs: active precordium, **soft pansystolic murmur or none (= LARGE defect)**, apical mid-diastolic murmur (↑mitral flow), **loud P2 (pulmonary hypertension)**. Cyanosis usually absent (unless PHT → Eisenmenger).
- **Investigations**: CXR (small = normal; large = cardiomegaly + pulmonary plethora/oedema). ECG (small = normal; large = **biventricular hypertrophy**). Echo = diagnostic.
- **Treatment**: small → reassurance + IE prophylaxis. Large → treat HF (**diuretics + ACE inhibitors**) + **surgical closure**.

Patent Ductus Arteriosus (PDA)

- Ductus connects pulmonary artery → descending aorta; normally closes after birth. PDA = failed closure by **1 month** post-birth. Common in **prematures**. **6–8% of CHD; F:M 2:1**.
- **Small PDA**: asymptomatic, normal heart size.
- **Large PDA**: symptomatic (HF, growth retardation), **bounding peripheral pulses, wide pulse pressure**, cardiomegaly, **machinery continuous murmur (max 2nd left ICS)**, systolic thrill.
- **Investigations**: CXR (large = LV/LA cardiomegaly, ↑pulmonary markings). ECG (large = LV/biventricular hypertrophy). Echo = main test.
- **Complications**: **infective endocarditis** (small + large); HF + emboli (large). Spontaneous closure after infancy extremely rare.

- **Treatment:** transcatheter closure, or surgical ligation/division (before 1st year). [Preterm: indomethacin/ibuprofen.]

Transposition of Great Arteries (d-TGA)

- **5% of CHD.** Aorta arises from RV (anterior, right of PA); PA from LV → **2 parallel circulations. Survival depends on mixing** (patent foramen ovale + ductus + VSD). Half have VSD. **M:F 3:1; commonest CHD in infants of diabetic mothers.**
- **Clinical:** cyanosis is predominant + always present — life-threatening, presents **day 2 (ductal closure)**. Loud single S2; usually no murmur.
- **Investigations:** CXR = "egg-shaped heart", narrow mediastinum. ECG = normal neonatal RV dominance. Echo confirms.
- **Treatment:** PGE (prostaglandin) infusion to keep ductus open; balloon atrial septostomy (Rashkind); arterial switch (Jatene) = treatment of choice (first 2 weeks — delay → ↓LV mass). Mustard operation; PA banding if VSD.

Tetralogy of Fallot (TOF)

- **5–7% of CHD. 4 components:** 1. RV outflow obstruction (infundibular PS) · 2. VSD · 3. overriding aorta (dextroposition + septal override) · 4. RV hypertrophy.
- **Clinical:** mild obstruction → HF, no cyanosis at birth; severe → **neonatal cyanosis (PDA-dependent)**. Older children: cyanosis, clubbing, exertional dyspnoea, squatting, growth delay.
- **Hypercyanotic ("tet") spells:** paroxysmal cyanosis, hyperpnoea, restless, gasping; common in **first 2 years**, mornings/after crying; **murmur DECREASES** during spell; prolonged → severe hypoxia/acidosis, rarely fatal.
- **Signs:** loud harsh ejection systolic murmur at left sternal edge (from day 1); murmur shortens + cyanosis increases as obstruction worsens; clubbing in older children.
- **Investigations:** CXR = "boot-shaped" heart, oligoemic lungs, right-sided aortic arch (20–25%). ECG = RVH, right axis, P pulmonale. Echo diagnostic.
- **Treatment:** medical initially → **definitive surgery ~6 months** (close VSD + relieve RVOT obstruction). Very cyanosed neonate → **modified Blalock–Taussig shunt** (subclavian → PA). **Tet spells:** knee-chest position, morphine, IV propranolol, fluids, bicarbonate, O₂; severe → paralysis + ventilation.

High-yield one-liners

- ASD = wide **FIXED** split S2; secundum → device closure; primum/AVSD = superior axis.
- VSD = commonest CHD; loud murmur = small defect; large → HF + loud P2 + surgery.
- PDA = machinery murmur + bounding pulses + wide pulse pressure; transcatheter closure; preterm = indomethacin.
- TGA = cyanosis day 2, egg-shaped heart; prostaglandin + balloon septostomy + arterial switch (first 2 wks). Diabetic mother.
- TOF = PS + VSD + overriding aorta + RVH; boot-shaped heart; tet spells (murmur decreases); BT shunt then repair ~6mo; spells = knee-chest + morphine + propranolol.
- Duct-dependent cyanotic lesions → prostaglandin keeps PDA open.

CONGENITAL HEART DISEASE — Revision Table

Bold = high-yield. 8/1000 liveborn; VSD commonest. [bracket] = standard added.

ACYANOTIC (left-to-right shunt)			
Lesion	Signs / murmur	Investigations	Treatment
ASD (secundum commonest; primum = partial AVSD)	Wide & FIXED split S2, soft ejection systolic; primum → apical pansystolic (AV regurg)	CXR: RA/RV enlarge, ↑pulmonary. ECG: secundum = RBBB + RAD; AVSD = SUPERIOR axis. Echo mainstay	Secundum → device closure; primum/sinus venosus → surgery (age 3–5yr)
VSD (commonest CHD, 30%)	Small = loud pansystolic LLSE (loud = small), quiet P2. Large = soft/no murmur, loud P2, mid-diastolic apical, HF	CXR: large = cardiomegaly + plethora. ECG: large = biventricular hypertrophy. Echo dx	Small → reassure + IE prophylaxis. Large → diuretic + ACEi + surgical closure
PDA (6–8%, F:M 2:1, prematures)	Machinery continuous murmur (2nd L ICS), bounding pulses, wide pulse pressure, thrill	CXR: LV/LA cardiomegaly. ECG: LV/biventricular hypertrophy. Echo	Transcatheter closure / ligation (<1yr); [preterm = indomethacin/ibuprofen]. IE risk
CYANOTIC			
Lesion	Key features	Investigations	Treatment
TGA (d-TGA)	5% CHD; aorta from RV. Cyanosis ALWAYS, presents day 2 (ductal closure); loud single S2. Survival needs mixing (PFO/PDA/VSD). Diabetic mother; M:F 3:1	CXR: egg-shaped heart, narrow mediastinum. ECG normal neonatal RV dominance. Echo	PGE infusion (keep duct open) + balloon atrial septostomy (Rashkind) + arterial switch (Jatene, first 2 wks)
TOF (5–7%)	4: PS (RVOT obstruction) + VSD + overriding aorta + RVH. Cyanosis (severe = PDA-dependent), clubbing, squatting. Tet spells (murmur DECREASES), first 2 yrs	CXR: boot-shaped heart, oligoemic lungs, R aortic arch 20–25%. ECG: RVH, RAD, P pulmonale	Medical → definitive repair ~6mo (close VSD + relieve RVOT). Cyanosed neonate → modified Blalock-Taussig shunt. Spells: knee-chest + morphine + IV propranolol + fluids + bicarbonate